

Research

Ecological role of foraminifera as bioindicators of eutrophication and heavy metal pollution: a case study of Estany dels Ponts Lagoon (Mallorca, Spain)

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Abstract

The Estany dels Ponts is a coastal lagoon in northeastern Mallorca, Spain, currently undergoing a restoration project focused on cataloging its biota and characterizing the ecological and physicochemical conditions of its water column and sediments. This study analyzes the total foraminiferal assemblage in 16 sediment samples collected throughout the lagoon and investigates its relationship with nutrient and element concentrations, including heavy metals and rare earth elements (REEs). Foraminifera, as key bioindicators, were examined (2900 individuals belonging to 140 species) to assess their distribution patterns and environmental interactions. Despite significant anthropogenic modifications since the mid-twentieth century, the lagoon maintains a relatively good conservation status, exhibiting meso-eutrophic to eutrophic conditions, likely driven by wastewater infiltration during peak tourist seasons. These conditions influence sediment chemistry through organic matter deposition, favoring the dominance of tolerant species. The sediment composition reveals spatial variability in nutrient and heavy metal concentrations, with urban wastewater identified as the primary pollution source. Although most metal concentrations remain within safe thresholds, nickel levels exceed ecological limits, posing potential risks, particularly in the inner lagoon. The foraminiferal community, dominated by *Ammonia aberdoveyensis*, reflects adaptability to fluctuating salinity and temperature. Species diversity is comparable to other *Cymodocea nodosa*-dominated habitats. Despite the high organic matter content and presence of heavy metals in sediments, no significant pollution-related effects were detected, as no foraminiferal test deformities were observed.

Keywords Coastal lagoon · Foraminifera · Bioindicators · Mediterranean

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1 Introduction

The anthropic effect caused by uncontrolled tourist development can have significant impacts on the environment. A clear example is the alteration suffered by the coastal areas of the Balearic Islands since the 1960s. The impact of this change has not only affected the coastline but also the water quality and seagrass meadows, as reflected by the evolution of foraminiferal assemblages, which not only demonstrate the impact but also the positive effect of mitigation measures [1, 2].

Coastal lagoons are another ecosystem affected by uncontrolled tourism development. These habitats represent ecologically significant points due to their uniqueness and the fact that they host particularly adapted species. This ecological importance has been the reason for their protection within the framework of European legislation.

The inclusion of the Water Framework Directive [3] in European protective regulations, along with the subsequent development of several directives and laws, as well as national and regional regulations based on it, has sparked increased interest in studying the condition of natural water bodies and in seeking bioindicators for their monitoring [4]. The WFD aims to achieve a “good status” for all EU waters by setting clear objectives and deadlines for water protection and restoration. This directive emphasizes the use of ecological and chemical indicators to assess the health of aquatic ecosystems and ensures that water bodies are managed sustainably across Europe [5, 6].

In response to these regulatory developments, there is an increasing focus on identifying reliable bioindicators—organisms or biological responses that can signal changes in the environment—to monitor the quality of water bodies. Bioindicators are essential because they can provide early warnings of ecological stress, pollution, or other adverse conditions affecting aquatic ecosystems, or inform about some negative influence in the past [4].

One of these coastal lagoons is the so called Estany dels Ponts. This site, listed in the Balearic Inventory of Wetlands with the code MAZH06, features a coastal lagoon that evolved from a much more extensive lagoon complex, so called s’Albufera, before its transformation in the 19th century. In the 20th century, the Estany dels Ponts and much of the surrounding wetland underwent profound changes, drastically altering its functions and values as a wetland.

As part of a larger lagoon complex, the Estany dels Ponts is linked with other water bodies like the remnants of the old the Estany Gran and Cibollar Lagoon, as well as several smaller, less identifiable lagoons. While the Cibollar Lagoon, a deep coastal lagoon with meromictic characteristics, has been extensively studied due to its unique biological community [7–10], there is limited and often fragmented information on the biotic and abiotic components of the Estany dels Ponts [11–13]. The water body of the lagoon system receives inputs from torrents, primarily the Torrent de Sant Miquel, which form the watershed located west of the study area. These torrents collect and distribute rainfall, as well as a portion of the water originating from the Serra de Tramuntana. Rainwater infiltrates through an aquifer composed mainly of Quaternary silts and sands, along with Lias (Lower Jurassic) limestones and dolomites [14, 15].

Given its significant natural value and the ecosystem services it provides, recent studies of the Estany dels Ponts have become particularly timely due to the intense anthropogenic pressures affecting this system. Highlighting its importance, a recently approved restoration project for the wetland, led by the Alcúdia City Council in collaboration with WWF and the University of the Balearic Islands (UIB), and funded by NextGenerationEU (2023–2025), has been initiated. This project includes cataloging the aquatic biota present in the lagoon, mapping, and ecological characterization of the water body and its sediments in terms of physicochemical parameters.

Through this comprehensive evaluation of the Estany dels Ponts’ ecological status from 2019, new information is provided such a high-resolution bathymetric characterization of the lagoon, the distribution of plants and macroalgae and a preliminary environmental evaluation. However, a more detailed and consistent analysis is needed regarding the bioindicator organisms. Foraminifera are shelled protozoans that serve as valuable indicators of environmental conditions due to their sensitivity to changes in water quality and sediment composition [16–20]. Due to their well-documented ecologies and the known stress tolerances of several species, the analysis of foraminiferal assemblages has proven valuable for assessing and monitoring coastal and shelf environments [16, 19–26].

Thus, the present work aims to offer a comprehensive characterization of the foraminiferal association and its application, along with the analysis of physico-chemical parameters, to evaluate the ecological state of the Estany dels Ponts.

2 Materials and methods

2.1 Study area

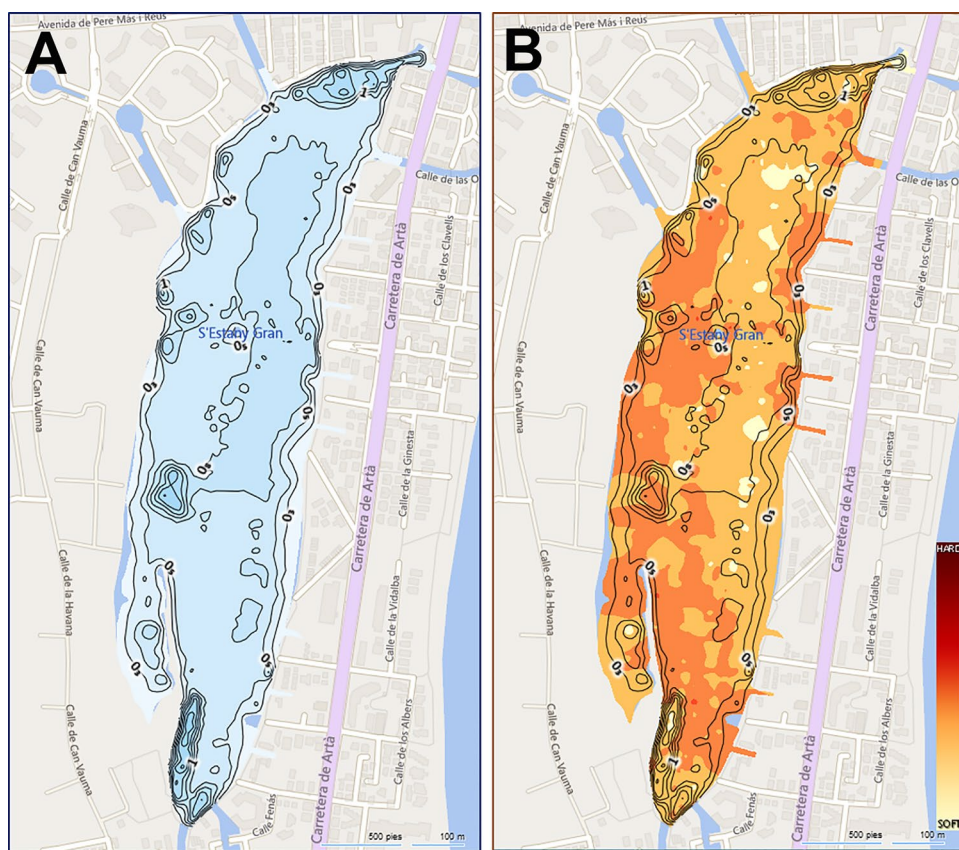
The Estany dels Ponts is a coastal lagoon placed on the NE of Mallorca (Balearic Islands, Spain). It is located adjacent to the dried-up the Estany Gran, near the town of Alcúdia, UTM EE-097085 (Fig. 1). The lagoon is connected to the sea in the north, while in the south, it is linked to the Canal des Cibollar, which leads directly to s'Oberta (Gran Canal) and, via a small diversion, to the Estany des Cibollar [11].

The Estany dels Ponts has a maximum length and width of 1515 m and 340 m, respectively, covering an area of 0.3911 km². The lagoon is enclosed by dune strips and displayed parallelly to the coast [11, 27]. The lagoon's morphometry has remained relatively unchanged since the late 1980's, maintaining a practically flat bottom with some very localized slopes (Fig. 2A). During a preliminary survey phase, the bathymetric characterization (Fig. 2A) of the lagoon



Fig. 1 Map with a general view of the sampling sites of the study area. **A** Map of Mallorca Island (Spain) in the Mediterranean Sea (arcgis.com, 2024). **B** Mallorca Island with the detailed location of the Parque Natural de s'Albufera (arcgis.com, 2024). **C** Map of the Estany dels Ponts (openstreetmap.org, 2024)

Fig. 2 Bathymetry maps (A) and bottom hardness (B) of the Estany dels Ponts. Available at <https://www.gesismaps.com/Dashboard/SocialMap>



was carried out using a Lowrance HOOK Reveal 5 sonar system equipped with a 50/200 HDI transducer. This instrument, in addition to providing depth data, allows for an approximate estimation of bottom hardness (Fig. 2B) based on signal reflectivity [28–30]. This approach establishes the association between hard bottoms and consolidated rock or coarse textures (gravels and coarse sands), and between soft bottoms and loose sediments with finer textures (medium to fine sands, silts, muds, etc.). Most of the lagoon is dominated by loose sediments, featuring considerable sand as well as silt and clay. These finer sediments, rich in silt and clay, characterize the two deepest zones, which exceed 2 meters, and are also present in various patches in the central part of the lake [31].

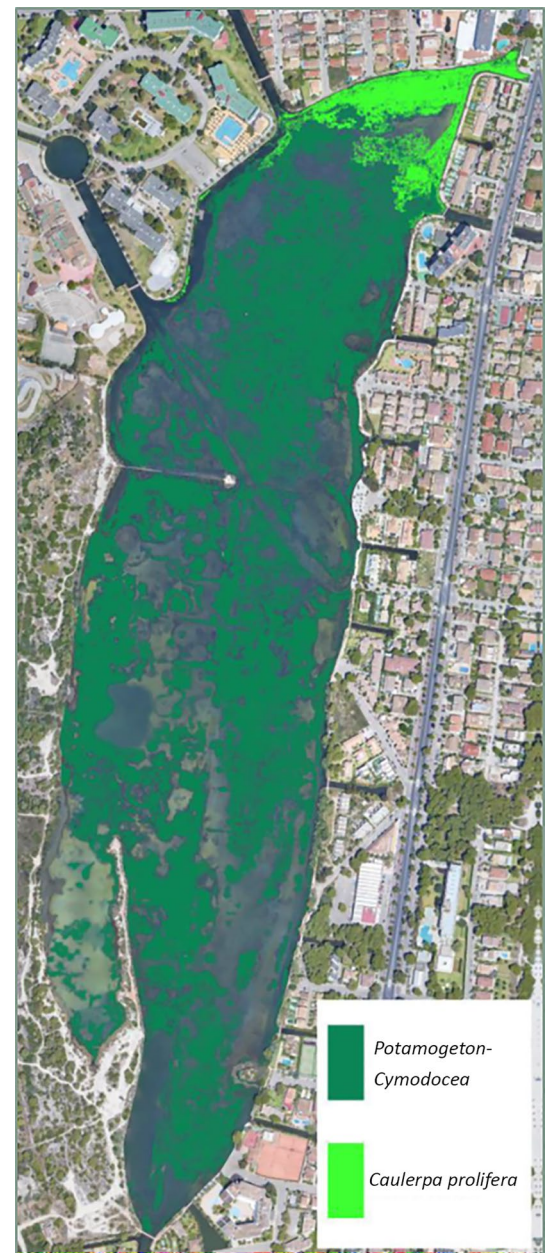
With an average depth of around 1 meter, the lagoon facilitates a significant exchange of gases with the atmosphere and supports the development of substantial benthic vegetation due to the high light exposure across most of the bottom. Depths exceed two meters in only two areas where the low light intensity reaching the bottom prevents vegetation growth. In a northern section of the lagoon, depths reach 1.5 meters, allowing the growth of macrophytes, specifically a community dominated by the green algal species *Caulerpa prolifera*, due to greater marine influence and light penetration in this area.

Due to its shallow depth, the Estany dels Ponts has a submerged macrophyte coverage exceeding 60% of its total area, with two very characteristic communities standing out (Fig. 3). According to Ferriol et al. [31] the area closest to the marine water inlet, towards the north, is dominated by a community of photophilous algae, mainly the large-sized *Caulerpa prolifera*, which hosts a high population of the anemone *Paranemonia cinerea*. Conversely, as the distance from the marine inlet increases towards the southern part of the lagoon, the *Caulerpa* community transitions into an interesting mixed community, composed of the marine seagrass *Cymodocea nodosa* along with the euryhaline submerged macrophyte *Potamogeton pectinatus*. This community exhibits very high densities, and in the case of *C. nodosa*, the length of the shoots is significantly greater than those found in coastal meadows.

2.2 Historical transformation

The Estany dels Ponts has undergone significant anthropogenic transformations over the years [32]. Originally part of a larger wetland system including several interconnected ponds (“estany” in Catalan). Historically, the Estany de Baix was

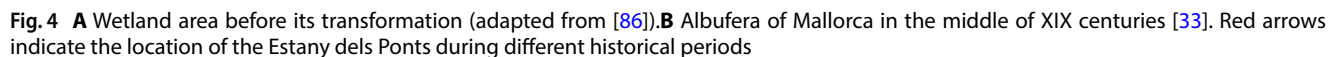
Fig. 3 Distribution of *Potamogeton pectinatus*-*Cymodocea nodosa* plant communities and the green macroalga *Caulerpa prolifera* on the bottom of the Estany dels Ponts lagoon



connected to the Estany dels Ponts by nine canals, of which only two remain today (Fig. 4). Under natural conditions, this complex was separated from the sea by a nearly-continuous dune barrier except for two main openings: a small canal at what is now called s'Oberta (Gran Canal opened in 1863), sometimes completely closed, and a significant opening at the Estany dels Ponts with an inlet over 200 meters wide. This setup allowed water to move slowly through the canals from one lagoon to another, eventually reaching the Estany dels Ponts and from there to the sea via the inlet.

The wetland area has been gradually and abruptly transformed both naturally and anthropologically. The first major intervention occurred in the 1860s with the drainage works of Albufera. Between 1865 and 1871, 2,146 hectares were drained to eliminate the unhealthy conditions caused by stagnant water and recover land for cultivation [32]. The construction of canals, ditches, and roads significantly altered the area's geomorphology and hydraulics, creating a new landscape marked by anthropogenic features. After the drainage, the land was primarily used for agriculture [33].

A significant infrastructure was required to manage production, leading to a more specialized agricultural exploitation from 1894 onwards. One notable use of Albufera's resources was the harvesting of reeds for paper production, with a paper mill operating from 1918 to 1966 [32]. In the late 1920s and early 1930s, the first tourists began to visit the area. Before the Spanish Civil War, the first chalets were built along Alcúdia's beach, along with the Golf Hotel and Mallorca's



Currently, the lagoon shore is urbanized in its northern half but remains recoverable in its southwestern half. The southern part of the lagoon's wetland area suffers from unchecked dumping of debris and garbage, and various unregulated activities, such as fishing and motocrossing [35]. The lagoon is also listed in the "Sports Facilities of the Center for Sports Technification of Palma de Mallorca" by the Higher Sports Council for canoeing [36].

In some environments, such as degraded seagrass meadows or locations of particular significance—like the site of this study—certain methodologies may inadvertently alter the habitat intended for protection. To address this issue, we

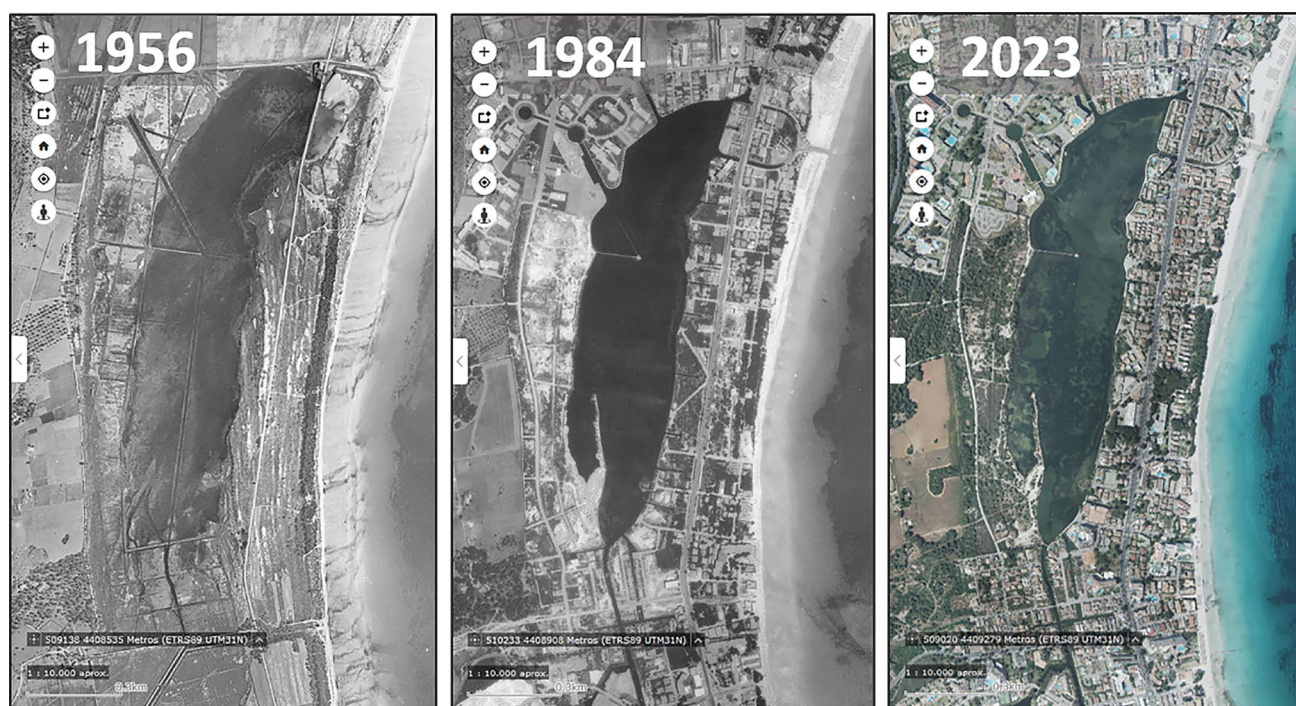


Fig. 5 Orthophotos of the Estany dels Ponts from 1956, 1984, and 2023 years (<https://ideib.caib.es/visor/?locale=es>)

adopted the total assemblage approach, as recommended in previous studies [19, 20, 38], which has been shown to yield valid results while significantly minimizing ecological impact. This methodology also allows for sampling foraminiferal species living on the surface and in the uppermost part of the substrate. To ensure that the sample is as representative as possible of in situ conditions and to avoid incorporating allochthonous specimens, only live foraminifera or those with pristine shells and no signs of reworking were considered.

To perform the analysis a subsample of each sample was used. The subsamples were initially treated with hydrogen peroxide solution for 24 h to eliminate organic matter attached to the foraminiferal tests. The sediments were thoroughly washed with tap water over a 0.063 mm mesh sieve and left to dry. Foraminifera were separated using a camera-equipped Leica MZ16 binocular stereomicroscope (camera Leica DFC295). Around one hundred and fifty foraminifera were identified per replicate sample. Taxonomic identifications were based on [39–45]. Nomenclature was updated according to the latest version of WoRMS [46].

2.4 Foraminiferal assemblage diversity (Shannon and Margalef) and bioindicator (Ep, FI, ILS, FSI) indices

Foraminiferal assemblage diversity was analyzed using the Shannon index (H') and the Margalef index (D_{mg}). The key difference between these indices lies in their approach to measuring diversity: H' considers both species richness (the number of species) and evenness (how individuals are distributed across species), while D_{mg} focuses primarily on species richness in relation to the total number of individuals in a sample. These indices were calculated using the PAST (PALaeontological STatistics) data analysis package.

Besides the taxonomic classification, foraminifera were analyzed according to their ecological preferences, particularly their epiphytic behaviour sensu Mateu-Vicens et al. [19] and their response to organic matter content [47]. To determine the abundance of epiphytic forms in the foraminiferal association, the Ep index (number of epiphytes relative to the total amount of individuals [19]) has been calculated for each sample. Those taxa with epiphytic behavior have been grouped under the different morphotypes representing life span, feeding strategy and motility [17, 42] (Fig. 7). The bioindicator indices were calculated based on ecological classification. The indices used in this study include the modified FORAM (Foraminifera in Reef Assessment and Monitoring) Index (FI'), the "Long vs. Short Life Span" Index (ILS), calculated by Khokhlova [48] and Mateu-Vicens et al. [19] using only epiphytic species, and the Foraminiferal Stress Index (FSI), as defined by Dimiza et al. [47] for the total foraminiferal assemblage.

Fig. 6 Maps of the sampling points of the Estany dels Ponts. Lines in the background represent isobaths

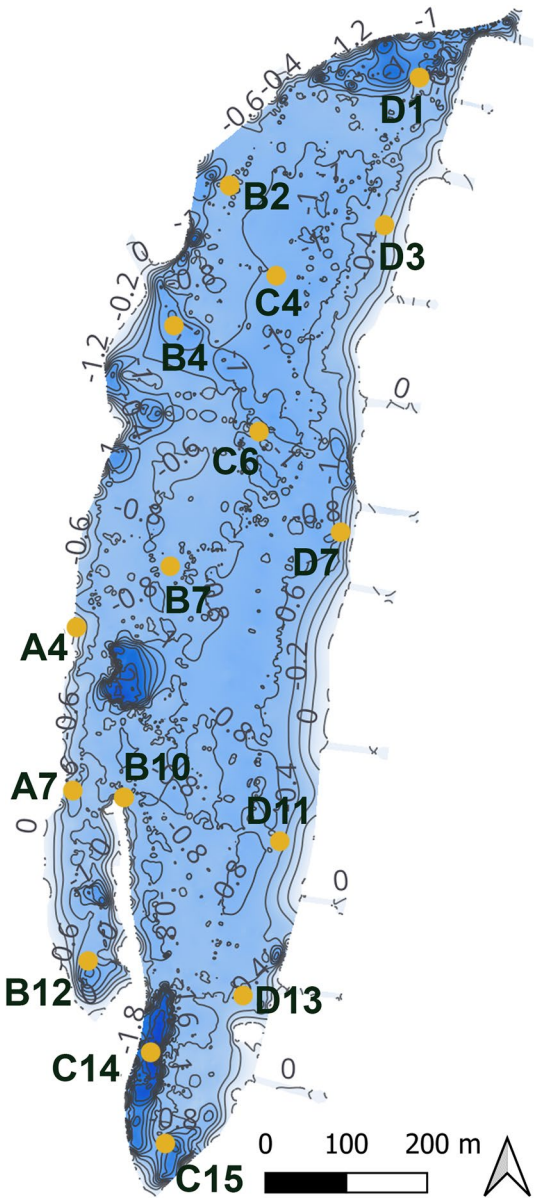


Fig. 7 Modified morphotype categories and their ecological features based on Langer [59], proposed in [48] and [19], where symbiont-bearing taxa are grouped into a new single morphotype SB (symbiont-bearing); renamed morphotype A* corresponds to long-lived, immobile taxa; and renamed morphotype D* corresponds to opportunistic taxa species; the morphotypes B and C correspond to the morphotypes of Langer [64] without changes

Khokhlova, 2013 Meteu-Vicens et al., 2014		Motility	Description	Life span
	A*	Encrusting	Mostly flat forms, permanently attached to the substrate by glycosaminoglycans. Ex: <i>Planorbulina</i> , <i>Nubecularia</i> , <i>Miniacina</i>	1 year
	SB	Sessile	Symbiont-bearing taxa. Ex: <i>Peneroplis</i> , <i>Sorites</i> , <i>Amphistegina</i> (Eastern Mediterranean)	1 year
	B	Temporary motile	Same as in Langer's (1993) classification	2-3 months
	C	Motile	Same as in Langer's (1993) classification	3-4 months
	D*	Permanently motile	Single aperture in upright position. Opportunistic forms. Ex: <i>Quinqueloculina</i> , <i>Triloculina</i> , <i>Adelosina</i> , <i>Textularia</i>	short

2.5 Sediment chemical analysis

Sediment samples were dried at 60 °C, homogenized, and divided into replicates. The organic carbon (OC) content of sediment samples was determined using a standardized calculation approach. Total carbon (TC) was measured via dry combustion, following the ISO 10694 method, in which sediment samples were combusted at high temperatures, and the resulting CO₂ was quantified using a thermal conductivity detector. Inorganic carbon (IC) was determined separately based on the ISO 10693 method, wherein sediment samples were treated with acid to release CO₂ from carbonates, and the carbonate content was calculated volumetrically. The organic carbon content was obtained by subtracting IC from TC. Additionally, sediment moisture was measured gravimetrically following ISO 11465, ensuring accurate sample preparation and weight correction. Total nitrogen (%) was analyzed using the persulfate digestion method [49] and quantified via the vanadium reduction colorimetric method [50]. Total phosphorus was measured following the method of Aspila et al. [51], after ignition at 550°C and treatment with 1N HCl, while inorganic phosphorus was determined using the same acid without ignition. Organic phosphorus was calculated as total phosphorus minus inorganic phosphorus. Samples were digested on an orbital shaker with HCl, centrifuged, and the supernatant was analyzed.

Subsequently, the C:N, N:P_{organic} and C:P_{organic} ratios were calculated to determine the C:N:P relationship or Redfield ratio [52], which will help to discern the origin of the organic matter in the lagoon.

2.6 Determination of chemical elements in sediments

Geochemical analysis was carried out on the sediments to determine the concentrations of major elements (typically > 1 wt% in rocks), trace elements (typically < 0.1 wt%, measured in ppm), and rare earth elements (REEs) (a subset of trace elements). The major elements in this study are Al and Fe; the trace elements, including heavy metals, are As, Cd, Cr, Cu, Mn, Ni, Pb, Zn, V, Co, Rb, In, Tl, Sc, Y. Among these, the REEs are represented by La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, and Lu. To perform the analysis the rest of the sediment samples were subjected to microwave digestion utilizing an ultra-wave oven (ETHOS 1, Milestone,) with 7 mL of HNO₃ (70% v/v) and 1.5 mL of H₂O₂ (30% v/v). All the elements were measured by Inductively Coupled Plasma-Mass Spectrometry (Agilent 7800), following a standard protocol [53]. The limit of quantification (LOQ) for trace elements was 0.010 mg.kg⁻¹ and for REEs 0.010 mg.kg⁻¹. The analytical methods were validated according to UNI CEI EN ISO/IEC 17025 (General Requirements for the Competence of Testing and Calibration Laboratories).

2.7 Statistical analysis

For statistical analysis Paleontological Statistics (PAST) software package v. 4.14 [54] was applied. Prior to conducting the analysis, the data were transformed using the log (X+1) function to reduce magnitude differences between variables. Two-way hierarchical clustering using the Ward's method and Euclidean distance measurement were also carried out on the datasets, in which Q-mode is used for clustering samples and R-mode for clustering variables (species) [55]. In this case R-mode refers to the clustering of species with abundances > 3% in at least on sample. To determine the robustness of the clusters, a cluster analysis with Bootstrap support (100 iterations) was performed separately in Q-mode and R-mode.

To determine if there is a correlation of the more abundant species from each sample, sediment parameters (organic carbon, total nitrogen, inorganic phosphorus, organic phosphorus), ecological (FI, I_{LS}, FSI), and biodiversity indices (Shannon H, Margalef) a Pearson analysis was performed. Principal Components Analysis (PCA) was calculated to determine the correlation between the spatial variation of the foraminiferal indices (FI, I_{LS}, FSI), biodiversity indices (Shannon H, Margalef), and sediment parameters (organic carbon, total nitrogen, inorganic phosphorus, organic phosphorus). The abovementioned diversity indices were also calculated using the PAST.

2.8 Mapping

Maps were created in QGIS to visualize the values of several parameters, including depth, total nitrogen, inorganic phosphorus, organic phosphorus and concentrations of major elements, trace elements and REEs in the sediment. For points where data were missing, the QGIS interpolation tool was employed to create continuous surfaces from discrete points.

3 Results

3.1 Nutrient concentration in sediments

The concentrations of organic carbon, total nitrogen, and organic phosphorus exhibit a similar distribution pattern (Fig. 8A, B, D; Table 1), with higher values in the northern half of the lagoon, including the samples along the north-south gradient (D1, B2, D3, C4, B4, C6, D7, and B7). However, particularly high concentrations of nitrogen and organic phosphorus stand out at C6 and D1, respectively. The far southern end of the lagoon, which includes sampling sites C14 and C15, also shows notably high concentrations of these nutrients, especially at C15, which coincides with the inflow point of channels carrying water from the Albufera wetland. In the southern half, excluding C14 and C15, concentrations are intermediate to low.

Regarding inorganic phosphorus (Fig. 8C; Table 1), its concentrations are low throughout most of the lagoon, with slightly higher values at the site closest to the connection with the sea (D1), along the western margin, and at the southern end (B12, C14, C15), particularly at D13. In the Redfield ratios calculated from the C:N, N:P_{org}, and C:P_{org} ratios, the C values range between 521 and 241, except for sample D13, which shows an unusually low value of 5. The N values range from 16 to 42.

3.2 Sediment geochemical analysis

The highest values of the elements concentrations are reported in the innermost part of the lagoon around B7 (see Table 2 and Fig. S1 in Supplementary information 1). Exceptions were found for Cu, with the highest concentration in the north around D1, and for Sc and Ti, with the highest values in the eastern part (D1, D7, D11, D13).

Fig. 8 Nutrient concentrations (%) in sediment of the Estany dels Ponts with interpolation data (QGIS). **A** Organic Carbon, **B** Total Nitrogen, **C** Inorganic Phosphorus, **D** Organic Phosphorus

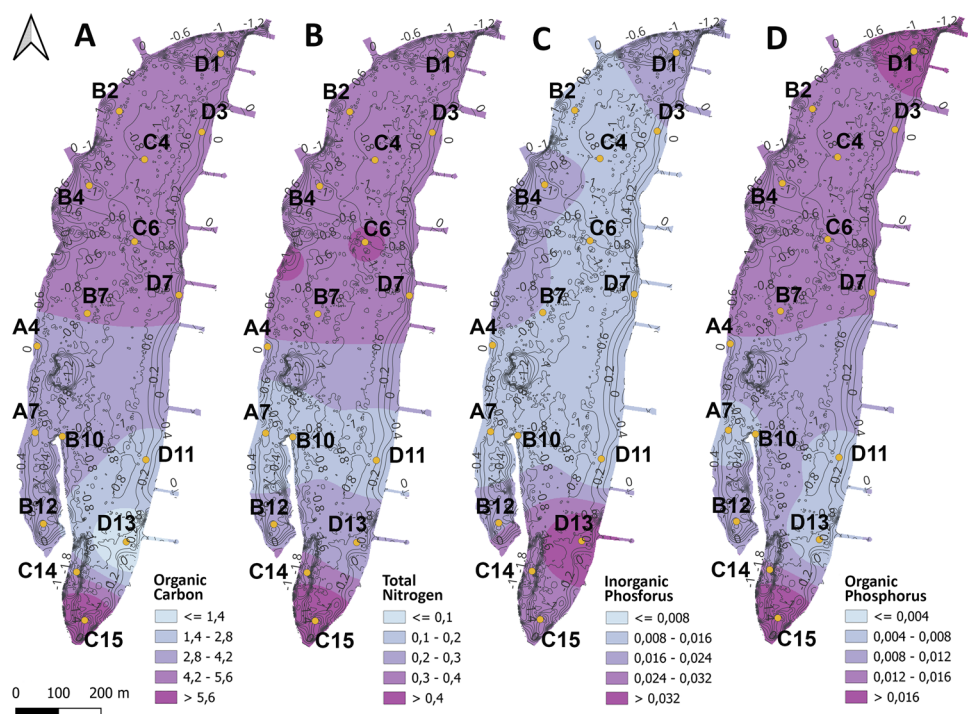


Table 1 Values of depth, organic carbon, nutrients (total N, inorganic P and organic P), nutrient ratios, foraminiferal indices (Index Ep, FI', I_{LS}, FSI), and biodiversity indices (Shannon H, Margalef) for the sampling sites of the Estany dels Ponts

Samples	Sediment	Foraminifera														
		Depth (m)	Organic carbon (%)	Total nitro- gen (%)	Inorganic phosphorus (%)	Organic phos- phorus (%)	C:N	N:P	C:P	Redfield ratio (C:N:P)	Index Ep	FI'	I _{LS}	FSI	Shannon H index	Margalef
A4	0.5	3.95	0.28	0.015	0.012	0.012	14.12	24.33	343.57	344:24:1	0.40	2.02	0.60	4.08	1.89	4.76
A7	0.5	3.65	0.11	0.008	0.007	0.007	33.18	15.71	521.43	521:16:1	0.28	1.72	0.27	2.87	1.80	5.92
B2	1.0	4.53	0.34	0.014	0.014	0.014	13.32	24.29	323.57	324:24:1	0.61	1.73	0.27	5.48	3.67	13.21
B4	1.0	4.58	0.32	0.017	0.015	0.015	14.31	21.33	305.33	305:21:1	0.25	3.00	2.17	2.90	1.38	4.36
B7	1.0	4.26	0.33	0.015	0.013	0.013	12.91	25.00	322.82	322:25:1	0.26	2.20	0.92	2.67	1.86	6.91
B10	1.0	3.52	0.13	0.012	0.008	0.008	27.11	16.83	456.22	456:16:1	0.55	4.27	3.42	5.25	2.81	6.40
B12	0.5	2.42	0.25	0.023	0.010	0.010	9.67	24.88	240.63	241:25:1	0.21	1.94	0.52	2.40	1.32	3.97
C4	1.0	4.62	0.33	0.014	0.015	0.015	14.01	22.66	317.56	318:23:1	0.12	1.67	0.23	1.88	0.72	2.58
C6	1.0	5.15	0.41	0.012	0.016	0.016	12.56	25.63	321.88	322:26:1	0.13	1.82	0.35	2.07	1.12	3.35
C14	2.0	3.91	0.35	0.030	0.013	0.013	11.16	26.39	294.55	295:26:1	0.26	2.04	0.63	3.22	1.34	3.39
C15	1.0	6.70	0.49	0.025	0.020	0.020	13.67	24.50	335.00	335:24:1	0.23	1.81	0.34	2.55	1.53	3.62
D1	2.0	5.31	0.37	0.017	0.017	0.017	14.35	21.76	312.35	312:22:1	0.73	3.42	4.54	5.80	4.05	14.98
D3	1.0	4.77	0.37	0.016	0.015	0.015	12.88	24.74	318.67	318:25:1	0.36	2.22	0.80	3.57	2.01	6.72
D7	1.0	4.50	0.35	0.013	0.013	0.013	12.85	26.58	341.66	342:27:1	0.67	2.20	0.92	6.41	3.04	7.57
D11	0.5	2.67	0.14	0.005	0.007	0.007	19.07	20.00	381.43	381:20:1	0.15	1.72	0.27	1.87	1.71	4.29
D13	0.5	0.03	0.25	0.036	0.006	0.006	0.12	41.67	5.00	5:42:1	0.81	2.07	0.62	7.26	3.55	13.42

The interpolated data using the QGIS program are indicated in italics

Table 2 Element concentrations in sediments (mg/kg) for each sample. Blue labels correspond to heavy metals and green labels represent REEs. Concentrations above the values of ERL of the NOAA are shown in red

	Al	As	Cd	Cr	Cu	Fe	Mn	Ni	Pb	Zn	V	Co	Rb	In	Tl	La	Ce	Pr	Nd	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Sc	Y
A4	4031	5.4	0.13	13.1	19.8	8160	157	84.7	17.04	138	12.37	1.22	12.41	0.043	0.16	6.4	10.0	1.4	6.2	1.3	0.3	1.5	0.23	1.20	0.24	0.70	0.086	0.52	0.076	3.86	14.76
A7	874	2.0	0.08	6.7	4.8	3694	138	1.6	2.01	10	3.85	0.44	2.93	0.003	0.10	5.3	7.6	1.2	5.3	1.1	0.3	1.4	0.20	1.03	0.21	0.61	0.074	0.45	0.065	2.34	15.81
B2	5710	8.7	0.24	17.6	27.9	7774	134	7.2	9.38	77	11.29	1.14	16.76	0.017	0.18	4.2	6.2	0.6	3.0	0.8	0.6	3.6	0.33	0.73	0.14	0.72	0.056	0.58	0.073	0.17	0.14
B4	4114	6.5	0.17	15.6	18.3	7863	142	5.4	7.68	51	8.92	0.98	11.74	0.011	0.13	6.5	9.8	1.4	6.4	1.4	0.4	1.7	0.25	1.25	0.25	0.75	0.090	0.57	0.082	2.12	6.85
B7	23633	15.0	0.62	44.0	21.4	11278	472	23.7	44.56	146	61.31	5.73	64.26	0.071	0.53	14.7	22.9	3.2	13.6	2.9	0.8	4.0	0.53	2.63	0.52	1.58	0.192	1.24	0.180	2.01	2.85
B10	5618	7.0	0.10	14.2	15.3	7835	173	5.6	8.29	30	22.33	1.22	15.88	0.012	0.13	5.9	9.0	1.4	6.0	1.3	0.3	1.9	0.25	1.21	0.24	0.75	0.087	0.57	0.083	1.50	2.98
B12	1937	2.5	0.14	6.8	15.6	6598	165	4.4	3.33	20	7.08	0.90	3.37	0.005	0.10	1.8	2.6	0.3	1.5	0.4	0.4	2.7	0.24	0.44	0.09	0.51	0.037	0.46	0.052	0.12	0.06
C4	10479	8.8	0.32	24.4	23.6	11609	295	13.7	19.71	64	38.71	2.98	30.09	0.025	0.26	7.9	11.9	1.7	7.7	1.7	0.5	2.9	0.35	1.56	0.31	1.00	0.112	0.79	0.112	0.84	1.15
C6	1658	2.2	0.09	6.1	14.3	6869	95	2.1	2.75	15	3.76	0.39	5.13	0.004	0.05	3.3	4.9	0.8	3.6	0.8	0.2	1.7	0.18	0.67	0.13	0.46	0.047	0.34	0.048	0.48	0.70
C14	7479	7.1	0.14	15.1	19.8	10533	151	7.1	14.67	63	18.86	1.51	24.64	0.018	0.17	1.4	2.0	0.2	1.1	0.4	0.9	3.8	0.33	0.47	0.10	0.66	0.046	0.65	0.065	0.06	0.02
C15	5589	4.5	0.21	11.5	12.0	8756	219	6.3	9.19	35	10.38	1.33	19.05	0.015	0.18	0.6	0.7	0.1	0.5	0.3	1.4	3.5	0.32	0.33	0.07	0.61	0.049	0.64	0.054	0.05	0.01
D1	2040	3.8	0.22	13.1	36.2	6656	70	7.1	7.18	62	4.25	0.56	6.78	0.009	0.12	5.2	7.2	1.1	5.3	1.1	0.3	1.2	0.16	1.07	0.20	0.57	0.072	0.48	0.062	3.57	35.44
D3	7079	9.3	0.25	22.0	34.6	10266	121	9.9	15.54	64	33.01	1.61	20.25	0.020	0.17	5.2	7.3	0.9	4.1	1.0	0.4	3.3	0.32	0.94	0.19	0.78	0.072	0.61	0.082	0.26	0.24
D7	9433	6.3	0.33	18.9	22.0	10530	188	9.7	23.05	72	23.59	2.15	34.65	0.030	0.33	7.2	10.9	1.6	7.1	1.4	0.3	1.5	0.19	1.20	0.21	0.61	0.077	0.52	0.067	4.74	14.74
D11	1993	2.7	0.10	7.3	6.4	4986	128	3.4	6.87	27	4.81	0.58	8.14	0.007	0.09	6.4	9.0	1.4	6.5	1.2	0.3	1.2	0.15	0.86	0.14	0.40	0.048	0.31	0.039	3.20	24.18
D13	926	2.1	0.06	6.2	7.4	4845	77	1.6	3.87	10	3.09	0.37	3.91	0.009	0.11	4.7	6.1	0.9	3.9	0.8	0.2	1.0	0.13	0.80	0.14	0.42	0.052	0.34	0.043	4.46	38.42
Min.	874	2.0	0.06	6.1	4.8	3694	70	1.6	2.01	10	3.09	0.37	2.93	0.003	0.05	0.6	0.7	0.1	0.5	0.3	0.2	1.0	0.13	0.33	0.07	0.40	0.037	0.31	0.039	0.05	0.01
Max.	23633	15.0	0.62	44.0	36.2	11609	472	84.7	44.56	146	61.31	5.73	64.26	0.071	0.53	14.7	22.9	3.2	13.6	2.9	1.4	4.0	0.53	2.63	0.52	1.58	0.192	1.24	0.180	4.74	38.42
Av.	5787	5.9	0.2	15.2	18.7	8016	170	12.1	12.19	55	16.73	1.44	17.50	0.019	0.18	5.4	8.0	1.1	5.1	1.1	0.5	2.3	0.26	1.02	0.20	0.69	0.075	0.57	0.074	1.85	9.90

The average heavy-metal concentrations from the the Estany dels Ponts are within natural ranges or below, according to the values of ERL (Effects Range-Low) of the National Oceanic and Atmospheric Administration (NOAA), which is consistent with low risk levels for most metals (Table 3).

However, there are moderate to high-risk values for nickel (Ni) concentrations in specific areas, such as B7 (23.7 mg/kg) and A4 (84.7 mg/kg), which exceed the ERM (Effects Range-Median) values, indicating a high likelihood of adverse effects on marine organisms in those regions. Arsenic (As), copper (Cu), manganese (Mn), and vanadium (V) present concentrations above the ERL but below the ERM, suggesting potentially moderate risks. For instance, arsenic values in B2 (8.7 mg/kg), B7 (15.0 mg/kg), and D3 (9.3 mg/kg) slightly exceed the reference for natural concentrations (8.2 mg/kg). Similarly, copper in D1 (36.2 mg/kg) and D3 (34.6 mg/kg), and for vanadium in B7 (61.31 mg/kg) slightly exceeded the reference concentrations for natural conditions. Manganese exhibited uniformly elevated values throughout the lagoon, ranging from 70 mg/kg to 472 mg/kg, with average moderately polluted sediment values ranging from 56 to 510 mg/kg. For the less common metals and rare earth elements, reference values are less established, and natural concentrations can vary significantly.

3.3 Foraminiferal assemblage analysis

Data collected from 16 points across different areas of the lagoon were analyzed to gather information about the species composition of epiphytic foraminifera. A total of almost 2900 individuals from 140 foraminiferal species, belonging to 28 families (Dataset_1.xls in Supplementary information and Supplementary Figure. S2), were identified in the total assemblage. Seventeen species occur with abundances over 3% in at least one sample and count for total abundances ranging between 52.85% (sample D1) and 96.62% (sample C14) (Table 4).

Foraminiferal assemblage in the lagoon is largely dominated by *Ammonia aberdoveyensis*, followed, at a much lower abundance, by *Lobatula lobatula*, *A. inflata*, *Cibicides refulgens*, *Rosalina obtusa*, *R. globularis*, and *R. floridana* among the calcareous perforate forms (hyaline), and *Quinqueloculina ungeriana*, *Triloculina marioni*, *T. trigonula*, *T. oblonga* and *Nubecularia lucifuga* among the porcelaneous forms (Table 4). *Ammonia aberdoveyensis* stands out as the principal species in most of the samples, comprising over 50% in 11 out of 16 samples (Table 4). This high abundance of *A. aberdoveyensis* is indicative of variable salinity and eventual low-oxygen periods [56]. The next most abundant species, *L. lobatula*, belongs to the epiphytic morphotype B. Morphotype C is represented by keeled species of genus *Elphidium*, *E. advena* (up to 4.72%) and *E. macellum* (up to 3.15%). Morphotype A, represented by genus *Nubecularia*, is the scarcest one, occurring only in 9 sites, and ranging, when present, up to 5.59%.

Table 3 The maximum (Max.) and average (Av.) heavy metals concentrations in sediments (mg/kg) for each sample

	Present study		Quality guidelines (SQGs)		NOAA		References	
	Max.	Av.	TEL	PEL	ERL	ERM	No contamin.	Contamin.
Al	23,633	5787					10,000	100,000
As	15.0	5.9	7.24	41.6	8.2	70		
Cd	0.6	0.2	0.68	4.21	1.2	9.6		
Cr	44.0	15.2	52.3	160	81	370		
Cu	36.2	18.7	18.7	108	34	270		
Fe	11,609	8016					20,000	50,000
Mn	472	170	56	510				
Ni	84.7	12.1	15.9	42.8	20.9	51.6		
Pb	44.56	12.19	30.2	112	46.7	218		
Zn	146	55	124	271	150	410		
V	61.31	16.73	60	130			20	50

The Sediment Quality Guidelines (SQGs) with two primary sets of criteria: TEL (Threshold Effect Level) and PEL (Probable Effect Level). National Oceanic and Atmospheric Administration (NOAA) with ERL (Effects Range-Low) and ERM (Effects Range-Median) values

The FI', I_{LS}, FSI, Shannon H, and Margalef indices were calculated for each sample considering the whole assemblage and not only the most abundant taxa (Table 1).

The Index Ep ranged from 0.81 in sample D13 to 0.12 in sample C4. Epiphytic foraminifers predominated (Ep > 0.50) in samples B2, B10, D1, D7, and D13. The FI' and ILS values ranged from 4.27 (B10) to 1.67 (C4) for FI' and from 4.54 (D1) to 0.23 (C4) for ILS. The FSI values varied significantly, from 1.87 (D11) to 7.26 (D13). Notably, the lowest and highest values of the Shannon H and Margalef indices coincided, with values of 0.72 and 2.58 in C4 and 4.05 and 14.98 in D1, respectively.

3.4 Data analysis

The Pearson correlation analysis revealed significant relationships among various species and indices (Fig. 9). In the foraminiferal assemblage, *A. aberdoveyensis* exhibits significant negative correlations with *A. parkinsoniana*, *As. mamilla*, *C. refulgens*, *E. macellum*, *T. marioni*, and *N. lucifuga*, as well as with the Ep, FSI, Shannon (H), and Margalef indices. In contrast, *E. macellum* shows a strong, significant positive correlation with *As. mamilla*, *C. refulgens*, *E. advena*, *R. globularis*, *N. lucifuga*, and with the Ep, FSI, Shannon (H), and Margalef indices. Moreover, the FI' and ILS indices exhibit a statistically significant positive correlation with *E. advena*, *R. globularis*, *R. obtusa*, and *N. lucifuga*.

Regarding nutrients, organic carbon, total nitrogen, and organic phosphorus all display a statistically significant positive correlation with one another. Organic carbon is negatively correlated with *N. lucifuga*, whereas total nitrogen is negatively correlated with *Q. ungeriana*. Inorganic phosphorus is negatively correlated with *M. affinis* and positively correlated with *N. lucifuga*. Depth is positively correlated with total nitrogen and organic phosphorus.

The cluster analysis performed on the samples (Q-mode) identified two main groups (Cluster A and Cluster B) with high robustness, as indicated by Bootstrap values of 100% and a Cophenetic Correlation Coefficient (Cophen. Corr.) of 0.7322 (Fig. 10). Cluster A is further divided into two subgroups; however, the low Bootstrap values (15%) suggest that this subdivision lacks robustness. The samples within each cluster exhibit comparable depth values and nutrient concentrations. However, Cluster B consists of samples B2, B10, D7, D1, and D13, which are characterized by the highest values in the Ep, FSI, Shannon H, and Margalef indices.

In the R-mode analysis, species are grouped into three highly robust clusters (Fig. 10). The separation of Group 1 from the rest and Group 2 from Group 3 is strongly supported by Bootstrap values of 100% and 94%, respectively. The Cophenetic Correlation Coefficient is 0.7641.

Cluster 1 consists exclusively of *Ammonia aberdoveyensis*, the dominant species in all samples, particularly in those forming Cluster A. Cluster 2 comprises a mix of epiphytic species (*T. oblonga*, *R. globularis*, *T. trigonula*, *T. marioni*, *A. mamilla*, *N. lucifuga*, *E. macellum*) and sediment-associated species (*A. parkinsoniana*, *A. beccarii*, *A. inflata*, *M. affinis*, *H. canariensis*). As shown in the heatmap, the species in Cluster 2 are not abundant but are present in samples from Cluster B, while being nearly absent in Cluster A. Cluster 3 includes typically epiphytic species (*L. lobatula*, *E. advena*, *C. refulgens*,

Table 4 The matrix of species from the Estany dels Ponts study area with abundance > 3%

Test type	Order	Family	Species	B10	A4	A7	B4	B12	C4	C6	C14	C15	D3	D7	D11	B2	B7	D1	D13
A	Lituolida	Haplophragmoididae	<i>Haplophragmoides canariensis</i>	0.00	0.00	1.27	0.00	1.30	0.00	2.52	0.00	0.00	0.00	4.08	0.00	0.00	1.89	0.00	0.62
H	Rotaliida	Ammoniidae	<i>Ammonia beccarii</i>	0.79	0.65	0.00	0.00	0.00	0.00	0.00	0.00	1.38	0.00	0.68	2.24	3.11	0.63	0.81	0.00
			<i>Ammonia inflata</i>	9.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.07	0.65	0.00	6.72	7.45	5.03	5.69	0.00
			<i>Ammonia parkinsoniana</i>	0.79	0.00	0.64	0.64	0.00	0.00	0.63	0.00	0.69	0.65	0.68	1.49	3.73	3.14	0.81	1.86
			<i>Ammonia</i> sp.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.38	0.00	0.00	4.48	3.11	0.00	2.44	0.00
			<i>Ammonia aberdoveyensis</i>	33.07	59.09	65.61	73.72	75.32	87.58	79.25	72.30	67.59	61.69	25.85	64.18	14.91	59.75	9.76	16.15
		Asterigerinatidae	<i>Asterigerinata mamilla</i>	0.79	3.25	1.27	0.00	0.00	0.65	0.00	1.35	1.38	0.00	2.72	0.00	0.62	0.63	4.07	3.73
		Cibicididae	<i>Cibicides refulgens</i>	4.72	1.95	0.00	0.64	1.30	0.65	1.26	0.68	6.21	2.60	4.76	0.75	2.48	3.14	4.88	1.86
			<i>Lobatula lobatula</i>	4.72	11.04	8.28	4.49	4.55	2.61	3.14	8.11	5.52	9.74	17.01	0.75	4.35	5.03	4.88	13.66
		Elphidiidae	<i>Elphidium advena</i>	4.72	1.95	2.55	1.92	2.60	0.00	0.63	1.35	2.76	1.95	1.36	2.24	2.48	0.63	4.07	1.86
			<i>Elphidium macellum</i>	3.15	0.00	0.64	0.00	0.65	0.00	0.00	0.68	2.07	0.00	1.36	0.00	1.86	0.63	2.44	3.11
		Melonidae	<i>Melonis affinis</i>	0.00	0.00	3.82	0.00	0.00	0.65	3.77	0.68	2.76	0.00	2.04	3.73	1.24	1.89	0.81	0.00
		Rosalinidae	<i>Rosalina floridana</i>	0.00	3.90	1.27	3.85	1.30	1.31	2.52	4.73	1.38	1.30	6.12	0.75	1.86	1.26	2.44	4.35
			<i>Rosalina globularis</i>	4.72	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.24	0.00	0.00	0.62
			<i>Rosalina obtusa</i>	6.30	1.30	0.64	1.92	0.65	0.65	0.00	3.38	0.00	0.65	4.08	0.00	0.62	0.63	2.44	0.62
P	Miliolida	Hauerinidae	<i>Quinqueloculina ungeri-ana</i>	3.15	1.30	1.27	0.64	1.95	1.31	0.00	2.70	0.00	1.30	4.08	2.99	0.62	0.63	0.00	1.86
			<i>Triloculina marioni</i>	1.57	1.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.07	0.63	0.81	0.00
			<i>Triloculina oblonga</i>	3.15	0.00	0.64	0.00	1.30	0.00	0.63	0.00	0.00	0.65	0.00	0.00	0.62	0.63	0.00	0.62
			<i>Triloculina trigonula</i>	1.57	0.00	0.00	0.00	0.00	0.00	0.63	0.00	0.00	0.00	0.00	0.00	4.35	0.00	0.00	0.00
		Nubeculariidae	<i>Nubecularia lucifuga</i>	3.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.24	0.63	3.25	5.59
			<i>Nubecularia</i> sp.	0.00	0.00	0.00	0.00	0.65	0.00	0.00	0.68	0.00	1.30	0.00	0.00	0.00	0.00	3.25	0.00
		Total		85.83	85.71	87.90	87.82	91.56	95.42	94.97	96.62	95.17	82.47	74.83	90.30	63.98	86.79	52.85	56.52
		Other taxa		14.17	14.29	12.10	12.18	8.44	4.58	5.03	3.38	4.83	17.53	25.17	9.70	36.02	13.21	47.15	43.48

Test types are A, agglutinate; H, hyaline; P, porcelaneous

R. floridana, *Q. ungeriana*, *R. obtusa*), which occur in both Cluster A and Cluster B, although they are slightly more abundant in Cluster B.

The PCA (Principal Component Analysis) (Fig. 11) reveals that 84% of the data variance is explained by the first two principal components (PCs), with PC1 and PC2 accounting for 60.4% and 23.6% of the variance, respectively. PC1 is primarily influenced by indices Ep and Margalef's and to a lesser extent, by total FSI and Shannon H index. PC2 is mostly influenced by organic carbon, followed by I_{LS} , depth and FI'.

Samples D7 and B2 align along PC1 and show a positive correlation with the Ep, Margalef, FSI, and Shannon H indices, whereas D3, A4, B7, C14, C15, C6, and C4 correlate negatively. The distribution of the remaining samples is influenced by both PC1 and PC2. Samples A7, B12, and D11 are associated with low I_{LS} values, in contrast to B10 and D1. Notably, D1 also exhibits very high values for the Margalef index.

4 Discussion

This discussion synthesizes the nutrient and geochemical analysis in the sediment including heavy metal and REEs concentrations to provide a comprehensive understanding of environmental dynamics within the Estany dels Ponts lagoon and how it affects the composition and distribution of the foraminiferal assemblage.

4.1 Nutrient and elemental characteristics of sediments

The sediment composition of the Estany dels Ponts lagoon reveals significant spatial variability in nutrient and element concentrations. The spatial distribution of organic carbon (C), total nitrogen (N), and organic phosphorus (P) concentrations suggests a higher accumulation of organic matter in the northern half of the lagoon and in the southernmost sampling site (C15). This latter point receives water input from the complex network of channels and lagoons connected to the Albufera.

The calculated Redfield ratios (C:N:P) for all samples are substantially higher than those typically observed in marine phytoplankton (106:16:1) but slightly lower than the values reported for seagrasses (550:30:1) [57]. However, the obtained values closely resemble those recorded in aquatic macrophyte detritus (383–315:16–13:1) [58] and in bacterial

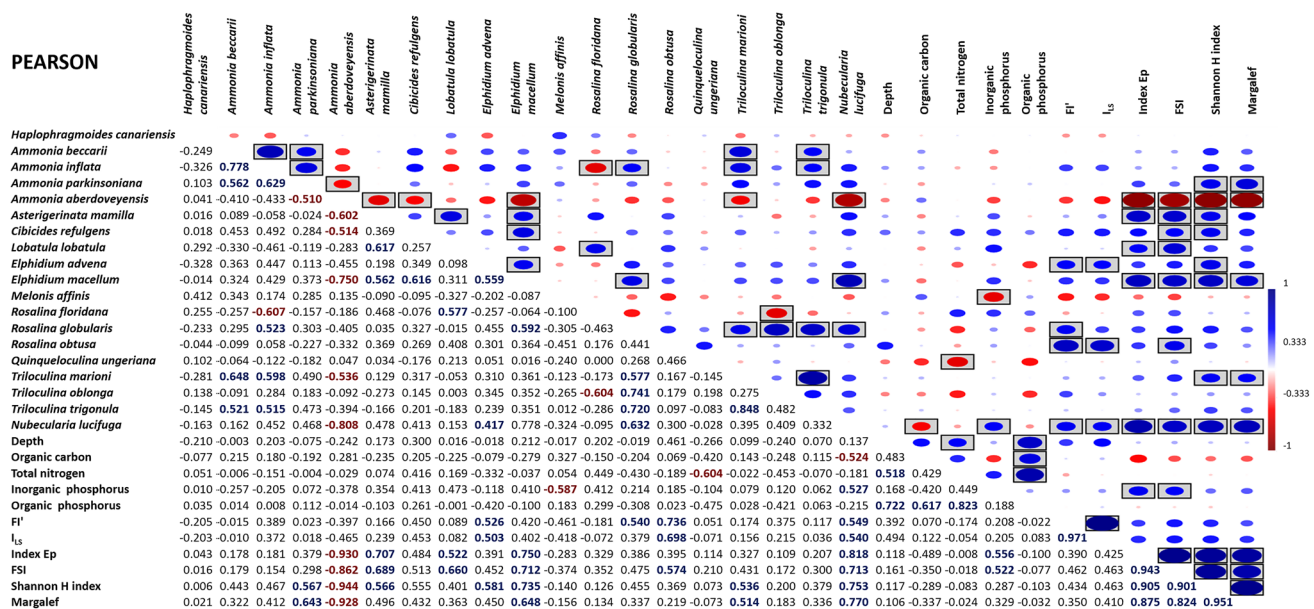


Fig. 9 Pearson correlation coefficients of most abundant species from each sample, depth, nutrients (total nitrogen, inorganic phosphorus, organic phosphorus), nutrient ratios, ecological indices (FI', I_{LS} , Ep, FSI), and biodiversity indices (Shannon H, Margalef). Blue ellipses represent positive correlations, while red ellipses represent negative correlations. The size of the ellipses indicates the r (coefficient) value. The grey boxes highlight significant correlations (p -value < 0.05)

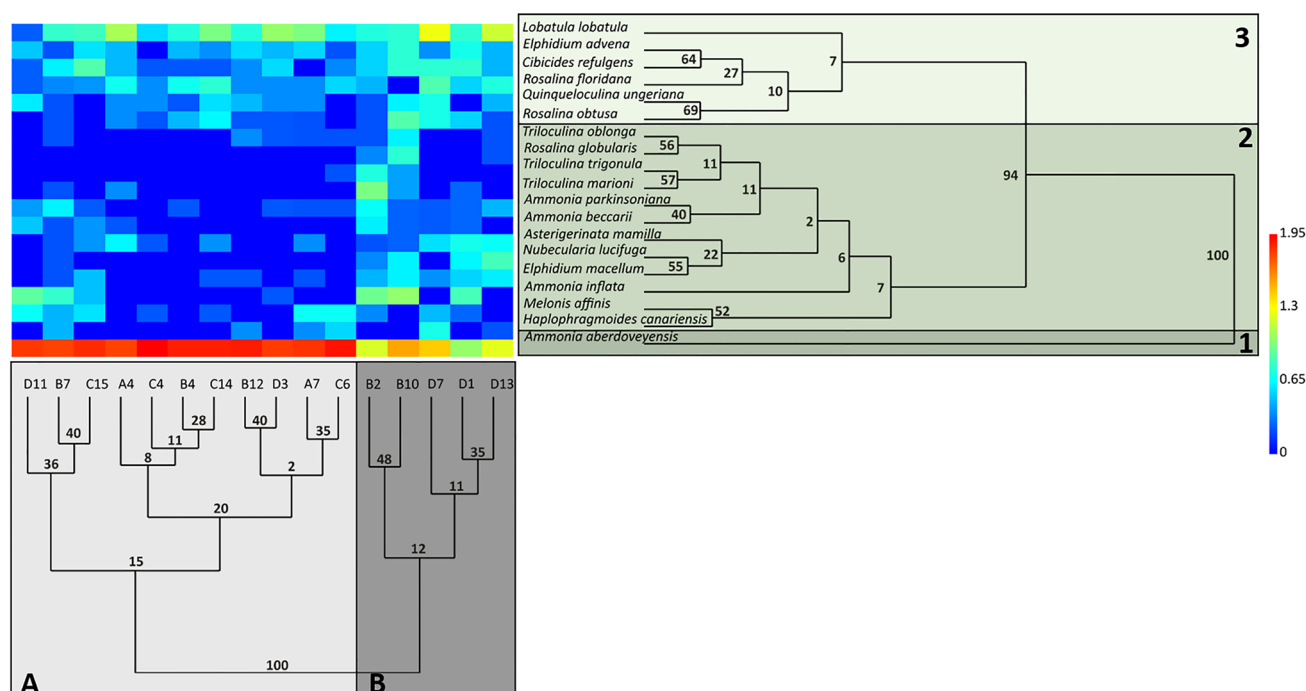


Fig. 10 Heat map and dendrograms generated from two-way hierarchical clustering analysis showing the distribution of foraminiferal genera in each sample through the lagoon the Estany dels Ponts. The heat map uses a color gradient to represent relative abundances, where red represents higher values and blue represents lower values. Both R- (species) and Q- (samples) mode clusters have been performed with bootstrap support (100 iterations). Numbers in the dendrograms are the bootstrap values. Cophenetic correlation coefficient (Cophen. Corr.): R-mode = 0.7641, Q-mode = 0.7322. Values close to 1 accurately reflect the relationships in the original data

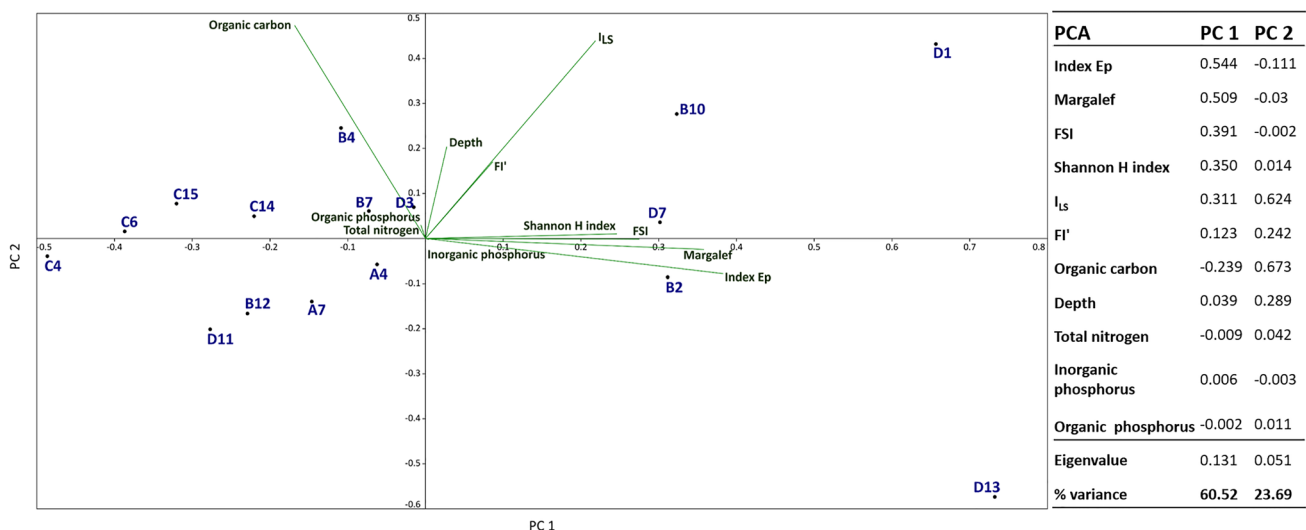


Fig. 11 PCA (Principal Component Analysis) plot for depth, nutrient concentrations, ecological indices (Ep, FI' , I_{LS} , FSI), and biodiversity indices (Shannon H, Margalef). The green line represents a variable (environmental or biodiversity indices). The direction and length of the vectors indicate the influence of the variable on the components, where the longer vectors have a stronger influence. Each blue point represents a sample from the lagoon. The position of the points indicates the relationship between the samples and the principal components. The table, on the right side, lists variables with their loadings (correlations) on PC1 and PC2, as well as the percentage of variance explained by each component

communities (320:34:1) from lakes with high water residence times [59]. These results likely reflect the combined contribution of different vegetation sources and bacterial decomposition processes in an environment constrained by phosphorus availability.

Sampling site D13 represents a notable deviation from the observed general trend. The calculated Redfield ratio (5:42:1) indicates low organic carbon content and an unusually high nitrogen concentration. The cause of this anomaly remains unknown, but it may be attributed to a localized disturbance, potentially linked to a point-source discharge or an external nitrogen input into the system.

Sediments serve as critical indicators of environmental contamination as they accumulate heavy metals over time, offering a historical record of pollution events [15]. While metals like Cr, Co, Cu, Fe, Mn, Mo, Ni, Se, and Zn are essential for biological functions and naturally occur in the environment, their excessive concentrations can become toxic [60]. Conversely, metals such as As, Sb, Cd, Pb, Hg, Ti, Ag, and Sn, which are not biologically essential, can signify anthropogenic contamination [61].

In the Estany dels Ponts lagoon, despite the absence of direct industrial discharges, heavy metal contamination has been largely influenced by decades of anthropogenic activities, such as urban wastewater and the discharge of ashes from power plants, as well as concrete and rubble [27]. Establishing baseline levels for these metals is critical for assessing the degree of pollution and its ecological impact [62].

Reference values for heavy metals, such as those provided by NOAA, are essential for identifying harmful concentrations. Sediment Quality Guidelines (SQGs), like TEL (Threshold Effect Level) and PEL (Probable Effect Level), help assess the likelihood of adverse effects on marine organisms [63]. The ERL (Effects Range-Low) and ERM (Effects Range-Median) values provide benchmarks to evaluate the potential ecological risks of metal concentrations. Heavy metal concentrations in the Estany dels Ponts lagoon vary significantly across different locations. The highest overall concentrations were found in the innermost part of the lagoon (B7), except for certain metals like Cu, which peaked in the northern region (D1), and Sc and Ti, which were more concentrated in the eastern samples (D1, D7, D11, D13). Most metal concentrations fell within natural ranges or below NOAA's ERL values, suggesting a generally low ecological risk. However, nickel (Ni) concentrations at stations B7 and A4 exceeded ERM values, indicating a high likelihood of adverse effects on marine life in these regions. Arsenic (As), copper (Cu), manganese (Mn), and vanadium (V) presented moderate risks, with concentrations above ERL but below ERM. These findings highlight the need for continued monitoring and mitigation strategies to address metal pollution, particularly in areas with elevated concentrations.

4.2 Foraminiferal assemblage and environmental conditions

The R-mode cluster analysis highlights the pronounced dominance of the *Ammonia aberdoveyensis* in the study area, as it exclusively forms Cluster 1. This suggests that *A. aberdoveyensis* is a key taxon, being dominant under the local environmental conditions that are characterized by seasonal variability mostly in temperature and salinity. In contrast, Clusters 2 and 3 comprise taxa commonly associated with an epiphytic lifestyle [19, 64–66]. This correlation is consistent with the widespread presence of vegetation across the bottom of Estany dels Ponts, supporting a habitat for epiphytic foraminifera. In fact, although cluster 2 includes species typically considered sediment dwellers (*Ammonia beccarii*, *A. inflata*), they exhibit an epiphytic behavior, at least seasonally [67]. In contrast, in Q-mode, cluster B includes the samples with the greatest influence from the epiphytic community and the best environmental indicators (higher diversity values and FSI scores).

Coastal lagoon environments, like the Estany dels Ponts lagoon, present highly variable conditions for benthic foraminifera due to intense seasonal and diel (day-night) fluctuations in temperature, salinity, and water quality. Such variability can be stressful for the foraminiferal assemblages [17], challenging their physiological limits and significantly influencing their distribution [19]. Despite these challenging conditions, the species richness (number of species) recorded in the Estany dels Ponts lagoon is comparable to that found in other seagrass habitats, such as *Posidonia oceanica* meadows in the Mediterranean [19, 68–71] and *Posidonia* species in Australia [38]. However, it is notably higher than that reported in *Halophila stipulacea* meadows in the Red Sea [72] and *Cymodocea*-dominated environments in the Mediterranean [73]. Despite variations in species richness across these studies, the diversity indices (H' and Margalef) from Estany dels Ponts are consistent with the ranges observed in similar environments. In contrast, lower diversity (H' values below 2) is observed in Lago Varano (SE Italy), where *Cymodocea nodosa* and *Nanozostera noltii* dominate [74].

There is limited literature describing and analyzing foraminiferal assemblages associated with Mediterranean *Cymodocea* meadows, especially when compared to the extensive research focused on *Posidonia oceanica*. One of the most detailed studies on *Cymodocea*-associated foraminiferal assemblages is by Blanc-Vernet [68], which identified important compositional similarities with *Posidonia oceanica* meadows, while also highlighting notable differences that give this assemblage a distinct character. In southern France, Blanc-Vernet observed that miliolids dominate these assemblages, followed by cibicidids, rosalinids-discorbids, and elphidiids. Other groups such as rotaliids, peneroplids, planorbulinids, nubeculariids, and nonionids occur less frequently, always constituting less than 10% of the assemblage.

In contrast, in the Estany dels Ponts lagoon, *A. aberdoveyensis* dominate (Table 4 and Dataset_1.xls in Supplementary information), while hauerinids (which correspond to miliolids in Blanc-Vernet's study) are less abundant. The abundance patterns of other taxa in the lagoon are consistent with Blanc-Vernet's observations. Blanc-Vernet suggested that the presence of homotrematids, planorbulinids, and nubeculariids indicated the influence of nearby *Posidonia oceanica* meadows. In the Estany dels Ponts lagoon, the presence of *N. lucifuga* among the more abundant taxa, and the encrusting species *P. mediterraneensis* and *Miniacina miniacea* (Morphotype A*, [19]), further underscores the influence of marine conditions, despite the lagoon's limited connection to the open sea. These species likely enter the lagoon through marine currents, high-energy events, or both.

The high salinities observed during the sampling period [31] in the Estany dels Ponts lagoon (41.0 PSU), linked to intense summer evaporation, can significantly influence foraminiferal distribution. This salinity measurement contrasts with previous reports from the same site, which included broader seasonal data with values as low as 24.12 PSU during winter and more humid months [75]. These salinity variability aligns with the dominance of the euryhaline species *A. aberdoveyensis*, known for their ability to tolerate wide salinity ranges [67, 76], and low occurrences of *A. beccarii*. This last species, although preferably associated with normal marine conditions, can also be found in habitats with salinities lower than 20 PSU [67]. The prevalence of *A. beccarii* has also been reported in other *Cymodocea*-dominated environments in the Mediterranean, with salinity levels reaching up to 39.1 PSU (Tunisia–[73]).

The dominance of *A. aberdoveyensis* points to its remarkable tolerance of environmental disturbances, including low oxygen [19, 56]. This prevalence suggests potential stress in parts of the lagoon, a conclusion supported by lower values of Ep, FSI and diversity indices. The negative correlation between this species and these indices highlights its ability to thrive in highly variable, stressful environments, where it outcompetes more sensitive, stenoic taxa.

The stressful conditions observed in the lagoon, characterized by high organic matter content and fluctuations in temperature and salinity, may be linked to both natural and anthropogenic factors. While human activities in the area cannot be disregarded, these conditions are also influenced by the natural organic matter inputs from its marine connection and the Albufera system. The inflow of marine water, along with brackish water contributions, combined with seasonal variations in evaporation and precipitation, drives the marked salinity fluctuations that define the lagoon's dynamic environmental conditions.

Interestingly, few specimens of mixotrophic species that host symbiotic microalgae, which are sensitive to excessive organic matter (morphotype SB, [19], were also observed distributed randomly along the lagoon. These taxa include *Sorites orbiculus* (D1), *Peneroplis pertusus* (B2, B4, B7, D1, D3), *Peneroplis planatus* (A4, D1), and *Laevipeneroplis proteus* (A4, B4, D13). Due to their relatively long lifespan (approximately 1 year–[64], these symbiont-bearing species require more stable conditions. Therefore, the observed shells might represent tests transported through the canal from *Posidonia oceanica* seagrasses in open marine conditions and subsequently redistributed throughout the lagoon by episodic high-energy hydrodynamic events (e.g., storms or strong winds).

However, the foraminiferal assemblage does not appear to show any significant negative effects from metal pollution or other environmental stressors, as no deformed tests were recorded. Benthic foraminifera, including epiphytic species, are considered excellent bioindicators of heavy metal contamination because they can incorporate specific trace elements (such as Cd^{2+} , Zn^{2+} , and Cu^{2+}) during calcification, which can result in various morphological abnormalities [17, 20, 25, 77–80]. However, the role of heavy metals in causing foraminiferal test deformations remains unresolved [18, 81], as deformed tests naturally occur in low percentages (<1%) in populations [22, 82, 83]. Additionally, a variety of natural environmental stressors—including high temperatures [84], low dissolved oxygen levels, salinity fluctuations (especially hypersalinity), periodic acidification, and test reconstruction after strong hydrodynamic events—can also contribute to abnormal growth forms [82, 83, 85].

5 Conclusions

Despite the significant anthropogenic modifications that the Estany dels Ponts lagoon has undergone, especially since the second half of the 20th century, its conservation status is reasonably good. Nevertheless, it presents meso-eutrophic to eutrophic conditions, characterized by high primary productivity and elevated nutrient concentrations, likely influenced by human activities, such as the leakage of domestic wastewater during the high tourist season. While these conditions promote productivity, they can also cause diel and seasonal fluctuations in oxygen, favoring tolerant species in the foraminiferal communities and altering sediment chemistry due to the deposition of organic matter.

Furthermore, the sediment composition in Estany dels Ponts shows significant spatial variability in nutrient and heavy metal concentrations, with pollution largely influenced by anthropogenic activities such as urban wastewater. Although most metal concentrations fall within natural or safe ranges, certain metals, such as nickel, exceed levels that could be harmful to lagoon life, particularly in the innermost parts and certain areas to the north and east. Therefore, it is important to establish baseline levels, monitor metal concentrations, and implement mitigation strategies to address potential ecological risks, especially in areas with elevated concentrations.

Foraminifera show a distribution influenced by a complex interaction of environmental factors. The Q-mode analysis indicates that samples associated with higher diversity indices and less stressful conditions are associated with higher abundances of epiphytic species. In contrast, the R-cluster analysis shows that the species *Ammonia aberdoveyensis* dominates the lagoon, suggesting that this species is key to the ecosystem due to its ability to adapt to environmental variability, especially in salinity and temperature.

Despite seasonal fluctuations, which favor tolerant species such as *A. aberdoveyensis*, species diversity in the lagoon is comparable to that in other habitats dominated by *Cymodocea nodosa*. Additionally, some mixotrophic foraminifera, typical of *Posidonia* meadows and requiring more stable conditions, are randomly distributed throughout the lagoon, suggesting that their shells may have been transported from nearby *Posidonia oceanica* areas through high-energy hydrodynamic events, such as storms.

As for the potential effects of metal pollution, no deformation was observed in the foraminifera, suggesting that there is no significant impact from metal pollution on the foraminiferal communities of the lagoon. Foraminifera are known bioindicators of metal pollution, but deformations in their shells can occur due to a variety of natural factors, such as fluctuations in temperature, dissolved oxygen, and salinity, complicating the direct link between heavy metals and shell anomalies. In summary, no significant negative effects from metal pollution seem to be present, and the observed species show remarkable adaptability to the fluctuating environmental conditions in the lagoon.

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Author contributions A.K. participated in the fieldwork, foraminiferal taxonomic identification, data analysis and writing. G.M.V. conducted and supervised the foraminiferal taxonomic analysis, contributed in the data analysis and wrote the manuscript. P.F.B. lead the field work, conducted the physical–chemical analysis and participated in the data analysis. S.S. and A.G. analyzed and interpreted the trace elements and the rare earth elements. E.B., L.C. and V.F. conducted the foraminiferal picking and taxonomic analysis.

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Data availability Data is provided within the manuscript or supplementary information files.

Declarations

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